MILITARY SPECIFICATION

SPLINT, LEG, PNEUMATIC, ADULT, WRAP-AROUND, HOOK AND CORD CLOSURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a full leg, double wall, plastic, pneumatic splint, inflatable by mouth.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-375 - Plastic, Film, Flexible, Vinyl Chloride.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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MILITARY

MIL-G-16491 - Grommet, Metallic.
MIL-F-21840 - Fastener Tapes, Hook and Pile, Synthetic.
MIL-C-43006 - Cloth And Strip Laminated, Vinyl Nylon High Strength, Flexible.
MIL-C-43307 - Cord, Nylon, Solid Braid, General Purpose.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)
25009 - Splint, Leg, Wrap-Around, Pneumatic Adult, Plastic

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be of the highest commercial quality and shall be entirely suitable for the purpose intended. Materials shall contain no ingredients known to be skin sensitizers or irritants.

3.1.1 Plastic.

3.1.1.1 Plastic for splint. **The plastic for the splint shall be vinyl film conforming to the requirements of L-P-375, Type I, Class 1.**

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3.1.2 Valve. The valve for opening and closing the air inflation passage shall be push-pull design. The exterior end of the valve stem shall be knurled for retention of the inflation tube. The valve housing and valve parts shall be fabricated from a polyvinyl chloride plastic suitable for the application. The valve shall be Halkey-Roberts, Inc. Part No. 330-AC, or equal.

3.1.3 Hook, Klondyke. The hooks shall be klondyke nickel plated steel boot hooks with design as shown on the drawing. Hooks shall be J. L. Thomson Rivet Co.'s Klondyke Hook No. 2, or equal.

3.1.4 Fastener, hook and pile. The hook and pile fastener shall conform to Type I or II of MIL-F-21840. Color shall be black.

3.1.5 Reinforcing Material. The reinforcing material shall be a light-weight laminated cloth composed of a nylon base fabric between two films of vinyl plastic. The fabric shall conform to the requirements of MIL-C-43006, Type I, Class 1 or 2.

3.1.5.1 Color. Color of the laminate shall be an intermediate shade (as pertains to light and dark) of gray, unless otherwise specified.

3.1.6 Cord, Nylon. The cord used for lacing the splint shall be braided nylon, 1/8 inch diameter, in accordance with MIL-C-43307. Length shall be 8 feet.

3.1.7 Grommet. The grommet used to anchor the cord shall be Type I of MIL-G-16491, size 00. Material shall be aluminum or brass.

3.1.8 Inflation tube. The inflation tube shall be fabricated from clear polyvinyl plastic material. The tube shall be flexible and shall not exhibit indications of plasticizer or additive migration such as a gummy or sticky surface, and shall not irritate the skin when used as intended.

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3.2 Design and construction. The design and construction of the leg splint shall be in accordance with Defense Personnel Support Center Drawing No. 25009. It shall be fabricated of two flexible layers of vinyl plastic (see 3.1.1), 0.020 ± .002 inches thick, fluted and sealed in such a manner as to form an air-tight envelope. In operation, the splint shall be closed and secured by lacing, utilizing a nylon cord and Klondyke boot hooks suitably placed to provide proper closure. Two airtight valves and an inflation tube shall be included to permit convenient inflation and deflation. The splint shall also have a hook and pile fastener, attached in such a manner as to form a pocket for the heel and foot when fastened.

3.2.1 Pressure test. The completed splint shall not show any reduction in air pressure (leakage) when tested as specified in 4.4.1.

3.3 Instructions for use. Instructions for use shall be furnished, either permanently and legibly marked on the plastic bag container or contained on a sheet and inserted inside the container. The instructions shall include appropriate illustrations. The instructions shall permit oral inflation only. A caution statement shall be included against the use of mechanical or hand pumps.

3.3.1 Government review of instructions. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed instructions for use, and Caution, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary text shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final text without delaying delivery under the contract or purchase order. Should the supplier have furnished acceptable instructions for use and

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and Caution for the identical item within two years preceding the date of contract and proposes to furnish instructions/ caution identical with those previously accepted, then the requirement for review of the draft may be waived upon request to the Contracting Officer.

3.4 Workmanship. Workmanship shall be first class throughout. Splints shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of Quality. Certificates of Quality supplied by the manufacturer of the plastic, hook and pile fasteners, reinforcing material, nylon cord and grommet, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, as shown in table I.

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-3, acceptance number shall be zero for all sample sizes, and a minimum of three samples shall be taken.

4.2.3 Component materials. Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the splints to perform tests as required by the applicable subsidiary specifications and by any tests specified herein. When sample size for component materials covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.2.2. The unit of product for testing shall be the quantity necessary to perform all required tests one time each. Where possible, the same material may be used to perform two or more tests. Lot sizes are expressed in multiples of the following:

Plastic	Yard or fraction of a yard
Laminated Cloth	Yard or fraction of a yard
Hardware	Each
Cord	Spool

4.3 Examination of end item. The splints shall be examined to determine compliance with all the requirements contained in this specification. The defects found during examination shall be classified in accordance with table I.

TABLE I. Inspection and Acceptable Quality Levels (expressed in defects per hundred units)

Examination paragraph	Inspection level	AQL Major	AQL Minor	Total
4.3.1	S-3	-	4.0	4.0
4.3.2	I	1.0	4.0	6.5

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4 3.1 Dimensional examination. Any dimension not within the specified tolerance shall be classified as a defect and classified as shown in Table I.

4 3 2 Visual examination Visual examination shall be conducted for defects in design, material, workmanship, construction and preparation for delivery Examination shall be conducted in accordance with but not restricted to the categories and defects listed in table II Additional defects shall be classified in accordance with MIL-STD-105.

TABLE II. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Any hole, cut or tear in the plastic.
102	Heat seal not continuous
103	Open seam.
104	Inflation tube missing.
105	Cord too large in diameter.
106	Cord missing.
107	Reinforcement for hooks missing
108	Tape fastener does not work properly
109	Plastic material not as specified
Minor	
201	Hooks improperly placed.
202	Grommet missing or misplaced
203	Tape fastener missing or attached improperly
204	Heat seal uneven.
205	Reinforcement for edge missing
206	Wrong type hooks
207	Tape fastener does not work properly
208	Wrong color of reinforcing material.
209	Finish of plastic wrong.
210	Marking omitted, misplaced, illegible, incorrect, or size of characters not as specified.

4 3 3 Packaging inspection The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements, as specified.

4.4.1 Pressure test. Sample splints shall be inflated to a pressure of 160 millimeters (mm) of mercury and the valve closed. The 160 mm pressure shall be maintained for one quarter-hour. The test shall be conducted at ambient temperature of 21°C to 25°C (70° to 77°F). Immersion in a water bath may be used to check for leaks and resulting pressure drop.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One splint, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Each splint shall be thoroughly clean, dry, neatly folded in a manner to prevent kinking or taking a permanent set and inserted into a plastic bag The splint shall be further packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676 Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each splint shall be clean, dry and individually packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A Ten unit packages shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

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5.2.2 Level B. Splints shall be packed as specified in 5.2.1 except that the exterior container shall conform to PPP-B-636, class domestic. In addition, waterproofing and strapping shall not be required.

5.2.3 Level C. The packaged splints shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129

6 NOTES

6.1 Intended use. The pneumatic splint is to be used in the first aid treatment of fractures of the leg and ankle.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.1 and 5.2).

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6.3 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6515-00-904-0116	SPLINT, LEG, Pneumatic, Adult, Wrap-Around, Hook and Cord Closure

6.4 Limitations. This specification does not cover all types of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity
(DLA-DM)

Project No. 6515-2032

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-37581 SPLINT, LEG, PNEUMATIC, ADULT, WRAP-AROUND, HOOK AND CORD CLOSURE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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